Dietary magnesium intake is inversely associated with mortality in adults at high cardiovascular disease risk.

The relation between dietary magnesium intake and cardiovascular disease (CVD) or mortality was evaluated in several prospective studies, but few of them have assessed the risk of all-cause mortality, which has never been evaluated in Mediterranean adults at high cardiovascular risk. The aim of this study was to assess the association between magnesium intake and CVD and mortality risk in a Mediterranean population at high cardiovascular risk with high average magnesium intake. The present study included 7216 men and women aged 55-80 y from the PREDIMED (Prevención con Dieta Mediterránea) study, a randomized clinical trial. Participants were assigned to 1 of 2 Mediterranean diets (supplemented with nuts or olive oil) or to a control diet (advice on a low-fat diet). Mortality was ascertained by linkage to the National Death Index and medical records. We fitted multivariable-adjusted Cox regressions to assess associations between baseline energy-adjusted tertiles of magnesium intake and relative risk of CVD and mortality. Multivariable analyses with generalized estimating equation models were used to assess the associations between yearly repeated measurements of magnesium intake and mortality. After a median follow-up of 4.8 y, 323 total deaths, 81 cardiovascular deaths, 130 cancer deaths, and 277 cardiovascular events occurred. Energy-adjusted baseline magnesium intake was inversely associated with cardiovascular, cancer, and all-cause mortality. Compared with lower consumers, individuals in the highest tertile of magnesium intake had a 34% reduction in mortality risk (HR: 0.66; 95% CI: 0.45, 0.95; $P < 0.01$). Dietary magnesium intake was inversely associated with mortality risk in Mediterranean individuals at high risk of CVD. This trial was registered at controlled-trials.com as ISRCTN35739639.